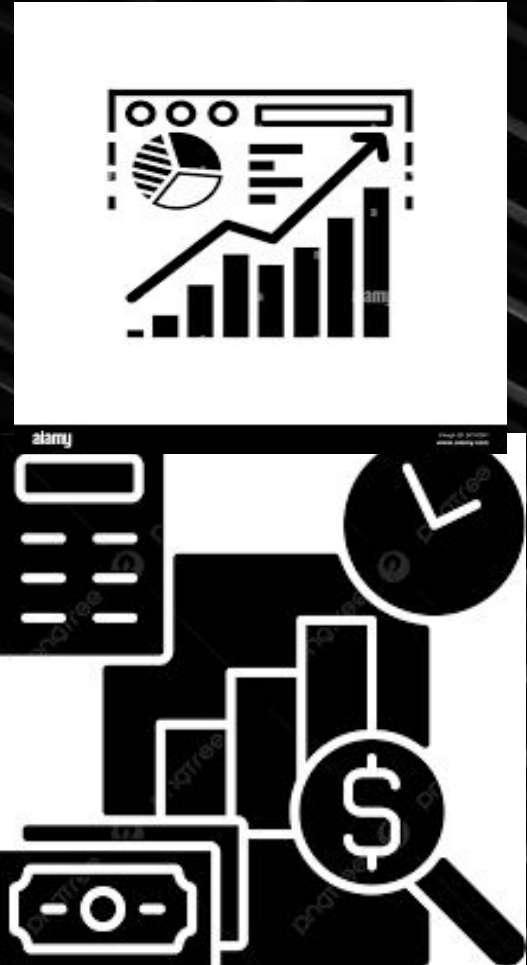


Financial Data Analysis with Pandas and SQL

By the Maryland Quantitative Finance Club

Introduction to Financial Data

- **Financial Data** powers quantitative analysis, allowing investors and analysts make informed decisions
- **Key Types**
 - Stock Prices
 - Fundamentals/Metrics
 - P/E, market cap, revenues
 - Market Indices - Performance data of markets
 - S&P 500, Nasdaq
 - Economic Indicators - Macro Data
 - Interest Rates, Inflation, CPI index & GDP





SCAN ME

Calculating Daily Returns & Log Returns with Python

```
df['Daily Return'] = df['Close'].pct_change()
```

Calculating the Daily
Return

Calculating Log Returns

```
df['Log Return'] = np.log(df['Close'] / df['Close'].shift(1))
```


Daily Returns and Log Returns

Daily Returns - The percentage change in stock price from one day to the next.

- tells you how much the price has grown or shrunk, as a percentage, over one day

Log Returns - found by taking the natural logarithm of the ratio between the new price and the old price of an asset

- a way to measure how much an asset's price has changed over a period, on a logarithmic scale

Volatility

Definition - measures how much a stock's price fluctuates (risk)

- Calculated by standard deviation of daily returns
- Higher volatility = higher uncertainty

```
volatility = df['Daily Return'].std()  
print(f"Stock Volatility: {volatility}")
```



Results and Visualization Analysis

Cumulative Return and Performance Evaluation

Definition:

Cumulative return tells you how much an investment has grown, or shrunk, over a period of time, **not just day-to-day changes..**

- How well did investment perform
- Answers the question of total gain
- Show how cumulative return **tracks long-term performance**, while daily return is **short-term noise**.

$$\text{Cumulative Return} = \frac{P_{\text{final}} - P_{\text{initial}}}{P_{\text{initial}}}$$

Cumulative Return Calculation

```
df['Cumulative Return'] = (1 + df['Daily Return']).cumprod()  
  
df[['Cumulative Return']].plot(figsize=(10,5), title="AAPL Cumulative Return")  
plt.show()
```

Risk Adjusted Performance: Sharpe Ratio



Definition:

The Sharpe Ratio analyzes how much return for each unit of risk. Indicator of high return worth.

Significance :

Helps compare strategies that may have similar returns, but different volatilities.

Two investments might give the same return, but if more volatile the risk is higher. Sharpe helps compare them fairly.

$$\text{SharpeRatio} = \frac{\text{Mean Return} - \text{Risk Free Rate}}{\text{Standard Deviation}}$$

Numerator: how much return above a safe investment (like U.S. Treasury bonds)

Denominator: how much investment's returns changed (volatility)

Interpretation:

- **Sharpe > 1** → good
- **Sharpe > 2** → very good
- **Sharpe < 1** → may not be worth the risk

QR Code

